

SODS Final Project - Group 26

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1 Baseline Synthesis

The baseline synthesis for the `sha1_core` design was performed using the `STcmos65` High- V_{th} (HVT) library. Through an iterative constraint-tightening process, the minimum clock period that satisfies both the setup timing constraints ($\text{Slack} \geq 0$) and the physical constraint ($\text{Area} \leq 24100 \mu m^2$) was identified as 5.5 ns. For the tighter constraint of 5.0 ns, the area exceeded the maximum allowed limit ($24229.22 \mu m^2$). Power metrics were accurately evaluated in Prime-Time using a VCD-based Switching Activity (SWA) back-annotation derived from post-synthesis functional simulation. The final extracted metrics are summarized in Table 1.

Table 1: Baseline synthesis results

Metric	Value
Clock period (ns)	5.50
Area (μm^2)	22835.56
Slack (ns)	0.0003
Leakage power (mW)	4.17×10^{-5}
Dynamic power (mW)	5.77

2 Power Optimization

2.1 Clock-Gating

Table 2: Clock-gating optimization results

Metric	Value
Area (μm^2)	21873.99
Slack (ns)	0.0013
Leakage power (mW)	4.14×10^{-5}
Dynamic power (mW)	6.17

Discussion. The introduction of clock-gating resulted in a reduction in overall area (from 22835 to 21874 μm^2) and a marginal improvement in the worst negative slack. The area reduction is primarily driven by the replacement of multiple data-path feedback multiplexers (typically used for register enable signals) with fewer Integrated Clock Gating (ICG) cells, according to the specified maximum fanout parameter. This architectural change also reduces the combinational logic depth on the data paths, slightly easing the setup timing constraints. However, dynamic power consumption slightly increased (from 5.77 mW to 6.17 mW). This counterintuitive behavior occurs because the SHA-1 core is subjected to continuous high-activity streaming in its functional testbench, meaning the enable signals are rarely de-asserted. Consequently, the design pays the switching power overhead of the added ICG cells within the clock tree, without fully reaping the benefits of disabling the registers. Conversely, leakage power decreased proportionally to the area reduction.

2.2 Clock-Gating + Multi-VT

Table 3: Clock-gating and Multi-VT optimization results

Metric	Value
Percentage of LVT cells (%)	21.48
Percentage of SVT cells (%)	0.05
Percentage of HVT cells (%)	78.47
Area (μm^2)	22564.01
Slack (ns)	0.0020
Leakage power (mW)	1.17×10^{-3}
Dynamic power (mW)	6.17

Discussion. The Multi-VT synthesis leveraged the available threshold voltage libraries to meet the constraints, resulting in a structural mix of approximately 78.47% HVT, 21.48% LVT, and a negligible fraction of SVT cells (0.05%). Compared to the purely HVT clock-gated baseline (Exercise 2.1), the introduction of LVT cells caused a significant and expected increase in leakage power (from 4.14×10^{-5} mW to 1.17×10^{-3} mW). This trade-off was necessary for the synthesis engine to resolve timing bottlenecks on critical paths introduced by the clock-gating logic and physical mapping, allowing the design to strictly meet the 5.5 ns clock period. The dynamic power remained practically unchanged (6.17 mW), confirming that the switching activity and the overall clock-gating efficiency were successfully preserved.

3 Custom Post-Synthesis Report

Table 4: Sample output of the custom timing report (Top 50 paths, max delay)

Cell Full Name	Cell Ref Name	Occurrences
add_3_root_add_0_root_add_315_4/U1_1	HS65_LH_FA1X4	1
add_1_root_add_0_root_add_315_4/U1_3	HS65_LH_FA1X4	1
add_1_root_add_0_root_add_315_4/U1_12	HS65_LH_FA1X4	1
add_1_root_add_0_root_add_315_4/U1_13	HS65_LH_FA1X4	1
add_1_root_add_0_root_add_315_4/U1_4	HS65_LH_FA1X4	1
add_1_root_add_0_root_add_315_4/U1_14	HS65_LH_FA1X9	1
add_3_root_add_0_root_add_315_4/U1_3	HS65_LHS1_FA1X9	1
add_1_root_add_0_root_add_315_4/U1_5	HS65_LH_FA1X4	1
add_1_root_add_0_root_add_315_4/U1_15	HS65_LH_FA1X9	1
add_1_root_add_0_root_add_315_4/U1_6	HS65_LH_FA1X4	1
...	...	...

Discussion. To validate the developed TCL procedure, the PrimeTime analysis script was executed, which automatically invokes the function `custom_report 50 max`. This configuration analyzes the 50 most critical paths for maximum delay (setup timing) and counts how many times each combinational cell appears within them. Table 4 shows an excerpt of the generated output. As requested, the script correctly filters out sequential elements and lists only the combinational logic gates, sorted in descending order based on their occurrence frequency across the analyzed paths.